

FORECASTING MARINE WAVE PATTERNS USING DEEP LEARNING TECHNIQUES WITH WAVE MONITORING DATA

Submitted by,

22BCE0985 SAPTAK SINGHA

22BCE2985 NVIZ YADAV

22BCE3100 ARITRA NANDA

Under the guidance of,

Dr. Mehfooza M.

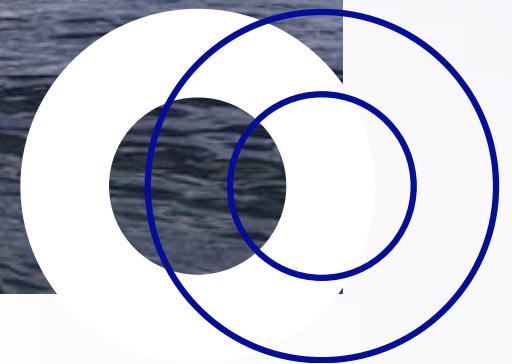
SYNOPSIS

This work explores deep learning approaches for multi-parameter ocean wave prediction using time-series data from ocean buoys. A comparative analysis of LSTM, GRU, and TFT models for multi-target ocean wave forecasting. Performance evaluation using RMSE, MAE, MAPE, and R^2 shows that TFT outperforms other models due to its attention-based architecture and superior handling of temporal dependencies.



INTRODUCTION

Oceanographic forecasting predicts the development of the sea state to assist in maritime navigation and coastal protection. Deep Learning models are implemented for forecasting wave parameters and the model with best performance is the chosen proving that it is effective in capturing temporal dependencies in oceanographic time-series data.



PROJECT GOALS

- **Need for Holistic Forecasting:** Most research focuses solely on predicting wave height. However, safe maritime operations require a multi-target approach that includes wave direction and period. This project is motivated by the challenge of building a model that predicts six inter-dependent variables simultaneously.
- **Overcoming RNN Limitations:** While LSTMs and GRUs are the standard for time-series, they often struggle with long-term dependencies^{[1][2]} and do not provide insights into which features (like SST or Direction) most influence the prediction.
- **Exploring State-of-the-Art (TFT):** The implementation of the Temporal Fusion Transformer (TFT) is motivated by its ability to provide interpretability through attention mechanisms and its advanced Variable Selection Networks, which help prioritize the most relevant oceanographic inputs for forecasting^[3].
- **Operational Safety:** Improving the accuracy of Hmax (Maximum Wave Height) and peak direction predictions directly contributes to reducing the risks of maritime accidents and coastal infrastructure damage.

References

[1] “Chang et al., "Dilated Recurrent Neural Networks," NeurIPS (NIPS) 2017.”

[2] “Vaswani et al., "Attention is All You Need," NeurIPS (NIPS) 2017.”

[3] “Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T., ‘Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting,’ International Journal of Forecasting, 2021.”

SCOPE

- **Data Analysis:** The study utilizes a dataset of approximately 43,454 records. It includes a robust preprocessing phase to handle missing sensor data (replacing -99.90 null values) and multi-target feature scaling.
- **Target Parameters:** The models are designed to predict a vector of six parameters:
 - Hs (Significant Wave Height)
 - Hmax (Maximum Wave Height)
 - Tz (Average Period)
 - Tp (Peak Period)
 - Peak Direction
 - SST (Sea Surface Temperature)
- **Model Architectures:** The scope includes the comparative implementation of three distinct deep learning frameworks: LSTM (multi-layered), GRU, and the Temporal Fusion Transformer (utilizing Multi-Head Attention and Gated Residual Networks).
- **Performance Benchmarking:** Evaluation is limited to point-forecasting on the test set, measured by RMSE, MAE, and R^2 scores to determine the most reliable model for practical maritime application.

PROBLEM STATEMENT

1. Complexity of Multi-Target Forecasting:

Most existing models focus on predicting a single variable (typically Significant Wave Height). However, for practical maritime operations, a single metric is insufficient.^{[1][2]} There is a critical need for models that can simultaneously and accurately predict a vector of correlated outputs, including wave height, direction, period, and temperature.

4. Data Quality and Noise: Raw buoy data is frequently plagued by sensor noise, outliers, and missing values (often represented as -99.9 in datasets). Developing a pipeline that remains robust under these conditions while extracting meaningful features is a significant technical hurdle.

2. Limitations of Traditional Numerical

Models: Traditional physical and numerical weather prediction (NWP) models are computationally expensive and often struggle to capture localized temporal patterns recorded by specific buoy stations in real-time.

5. Model Interpretability and Efficiency:

There is a trade-off between the simplicity of traditional RNNs and the high performance of modern Attention-based architectures (like Transformers). Determining which architecture provides the best balance of predictive accuracy and computational efficiency for multi-target oceanographic data remains a key research gap.

3. Capturing Long-Range Temporal

Dependencies: Standard time-series methods often fail to remember long-term historical patterns that influence future wave states. While Recurrent Neural Networks (RNNs) like LSTM and GRU have improved this, they often struggle with gradient stability and the ability to prioritize the most relevant historical time steps.^{[3][4]}

References

- [1] "Prediction of significant wave height based on EEMD and deep learning (Frontiers in Marine Science, 2023)"
- [2] "Using Machine Learning Techniques to Predict Significant Wave Height (Engineering and Applied Sciences, 2024)"
- [3] "Chang et al., "Dilated Recurrent Neural Networks," NeurIPS (NIPS) 2017."
- [4] "Vaswani et al., "Attention is All You Need," NeurIPS (NIPS) 2017."

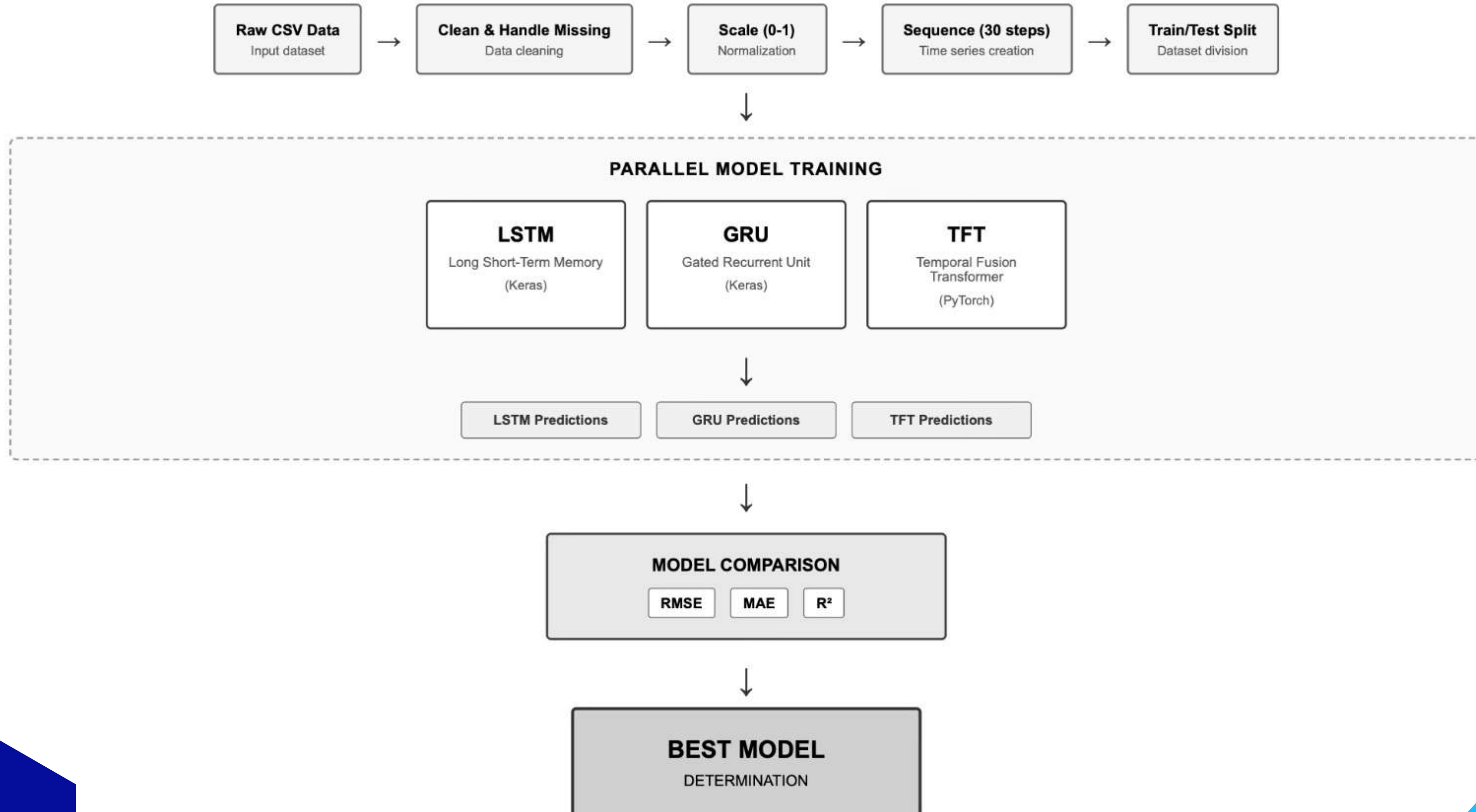
OBJECTIVES

- 1. Design a Multi-Model Deep Learning Framework:** To apply and train three different models (LSTM, GRU, and TFT) on the task of predicting the Significant Wave Height (SWH).
- 2. Assess the Handling of Long-Term Dependencies:** To compare the performance of the Gating mechanisms in RNNs (LSTM/GRU) and the Self-Attention mechanism in TFT on the task of capturing long-term wave patterns (e.g., swell waves).
- 3. Assess Model Interpretability:** To apply the attention weights of the TFT model to determine the most important past time steps and the influence of wind vs. wave history on extreme wave events.
- 4. Benchmarking Performance vs. Efficiency:** To measure the trade-off between the speed of GRUs and the accuracy of TFTs on the task of multi-horizon wave height prediction.

SYSTEM OVERVIEW

- **Data Layer:** Ingests buoy data (43k+ records) containing physical wave and thermal (SST) features. The system identifies and utilizes six primary oceanographic variables for both input and multi-target prediction:
- **Hs**(Significant Wave Height), **Hmax** (Maximum Wave Height), **Tz**(Average Wave Period), **Tp**(Peak Energy Wave Period), Peak Direction (Wave direction relative to True North), **SST**(Sea Surface Temperature)
- **Preprocessing:** Cleans sensor errors (-99.90 values) and normalizes features for neural network stability.
- **Modeling:**
 - LSTM/GRU: 3-layer recurrent architectures for temporal patterns.
 - TFT: Uses Multi-Head Attention and Variable Selection Networks for high-accuracy forecasting and feature importance.
- **Evaluation:** Multi-panel visualization comparing real vs. predicted values for all six targets.

SYSTEM DIAGRAM



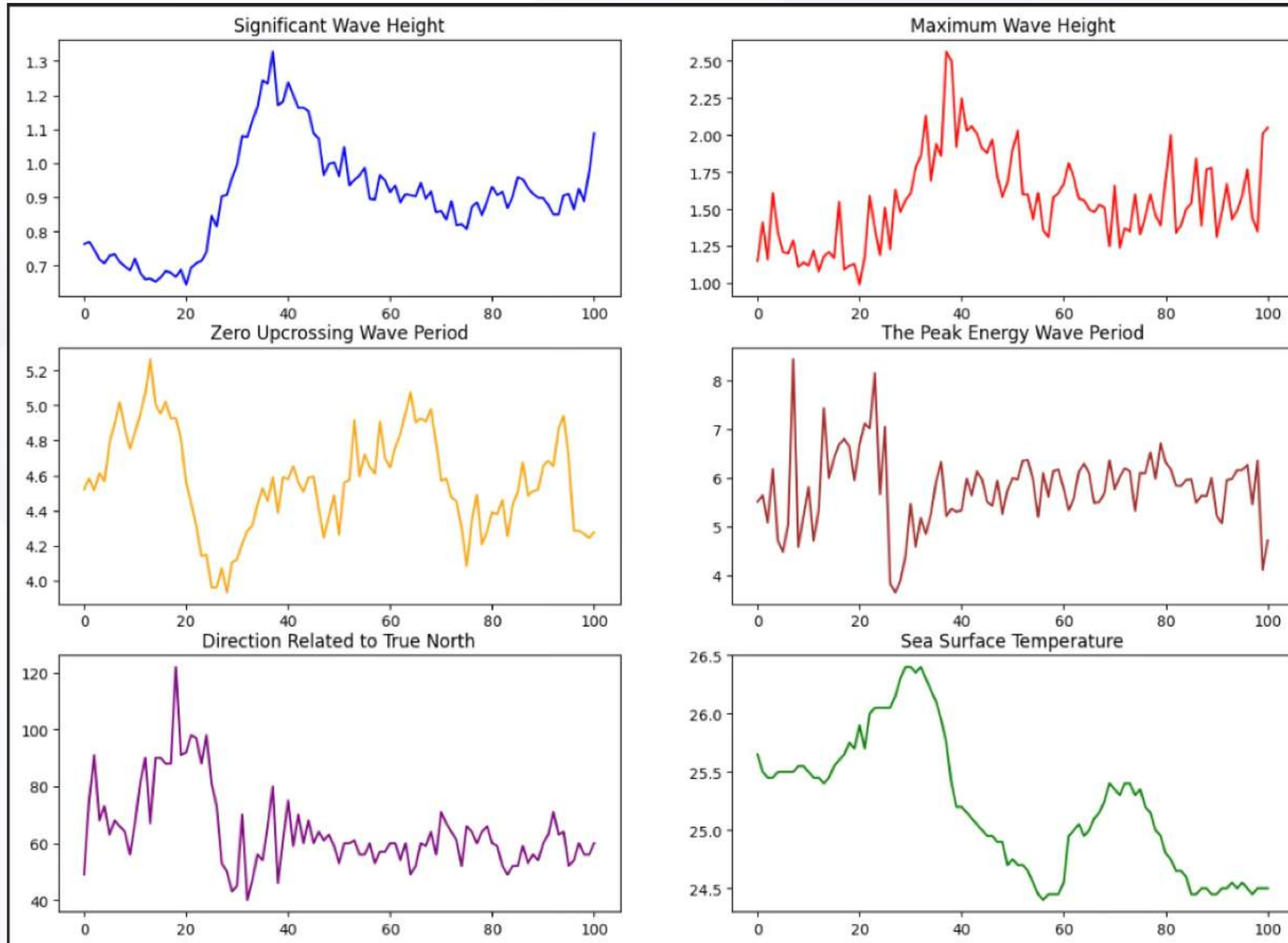
DATASET ACQUISITION

- It is Mooloolaba Coastal Wave Dataset available through the Queensland Government Open Data Portal:
<https://www.data.qld.gov.au/dataset/coastal-data-system-waves-mooloolaba>
- The dataset consists of certain measured and derived ocean wave parameters such as Significant Sea Surface Temperature (**SST**), Wave **Direction** (relative to true north), Peak Energy Wave Period (**T_p**), Zero Upcrossing Wave Period (**T_z**), Maximum Wave Height (**H_{max}**), Significant Wave Height (**H_s**) recorded at **30-minute intervals** using an offshore wave buoy monitoring system.

DATA PREPARATION

- All the missing value instances has been handled by replacing them with -99.90 and later NaN.
- Feature normalization was applied and features were scaled to a bounded range [0, 1] using Min-Max scaling.
- The normalized data was partitioned into discrete sequences using a sliding window technique.
- Dataset is partitioned into train and test sets with 42000 entries to train and the rest for testing.

VISUALIZATION OF THE FEATURES



TOOLS AND SOFTWARES

- **Primary IDE:** Visual Studio Code
- **Programming Language:** Python (version 3.8 or above)
- **Data Processing and Analysis Libraries:** NumPy (Numerical Computations), Pandas (Data Handling and Preprocessing)
- **Visualization Tools:** Matplotlib (Plotting and Result Visualization), Seaborn (Correlation heatmaps and statistical plots)
- **Machine Learning and Deep Learning Frameworks:** Scikit-learn (Scaling, Matrices, and Data Splitting), TensorFlow (Implementation of LSTM, GRU, TFT models)

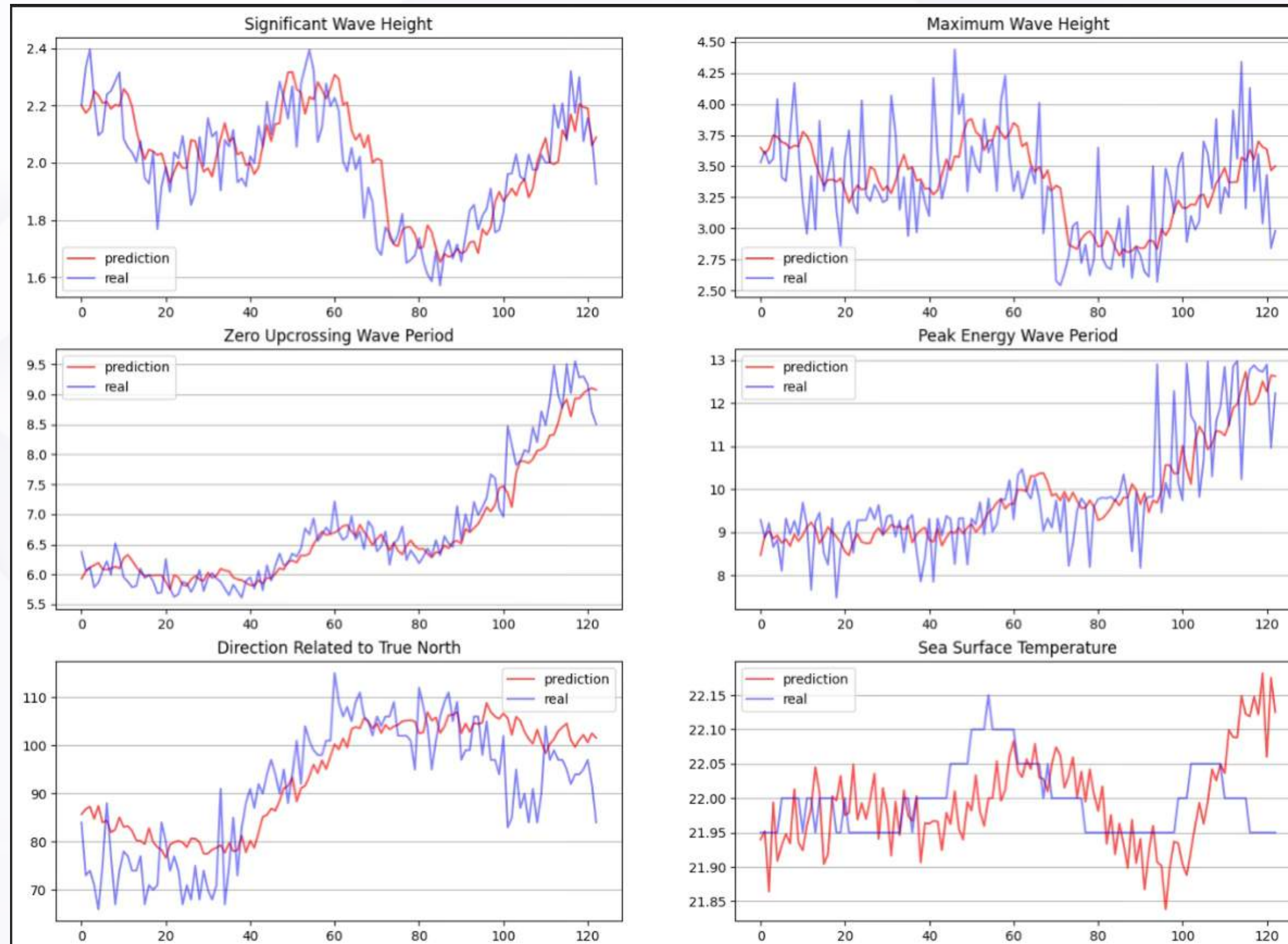
LSTM

- It reduces the vanishing gradient problem, very common in RNNs^[1].
- The model contains three consecutive LSTM layers with 32, 16, and 10 hidden units, respectively.
- An integration of a dropout layer having rate of 0.2 between second and third layers of LSTM was done to reduce the overfitting risk.
- Adam optimizer was employed with learning rate = 0.001

References

[1] "Hochreiter, S., & Schmidhuber, J., 'Long Short-Term Memory,' Neural Computation, 1997."

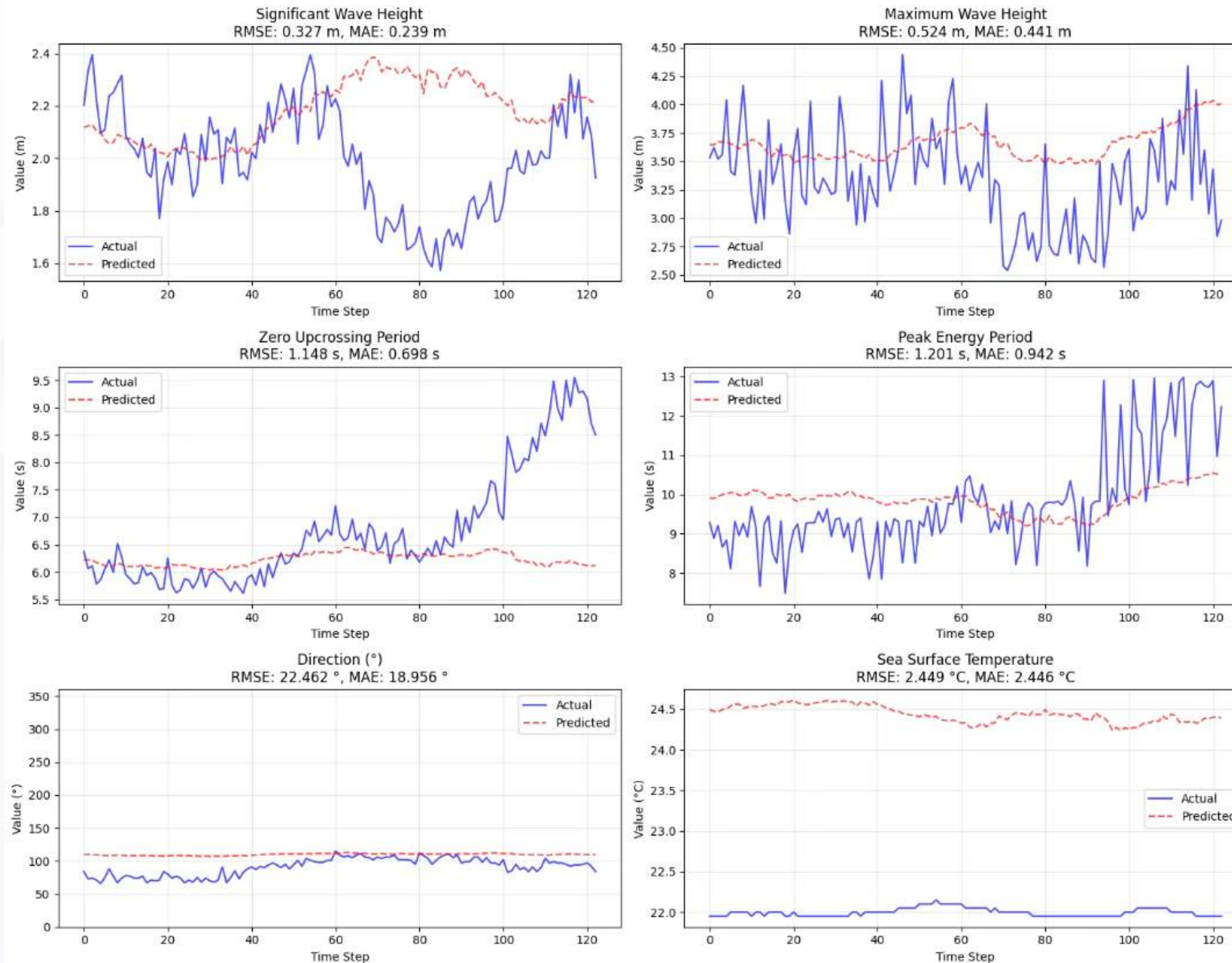
LSTM VISUALIZATION: REAL VS PREDICTED



GRU

- The proposed GRU architecture consists of a deep stacked configuration of (128 to 64 to 32) is for hierarchical feature extraction.
- Elastic net regularization was applied where $\lambda_1 - \lambda_2 = 0.001$ on both kernel and recurrent weights to avoid overfitting.
- Progressive dropout rates (30%, 30%, 20%, 20%) has been applied to improve generalization and prevention of co-adaptation of neurons.
- Adam optimizer was employed with learning rate = 0.0005, clipnorm = 1.0

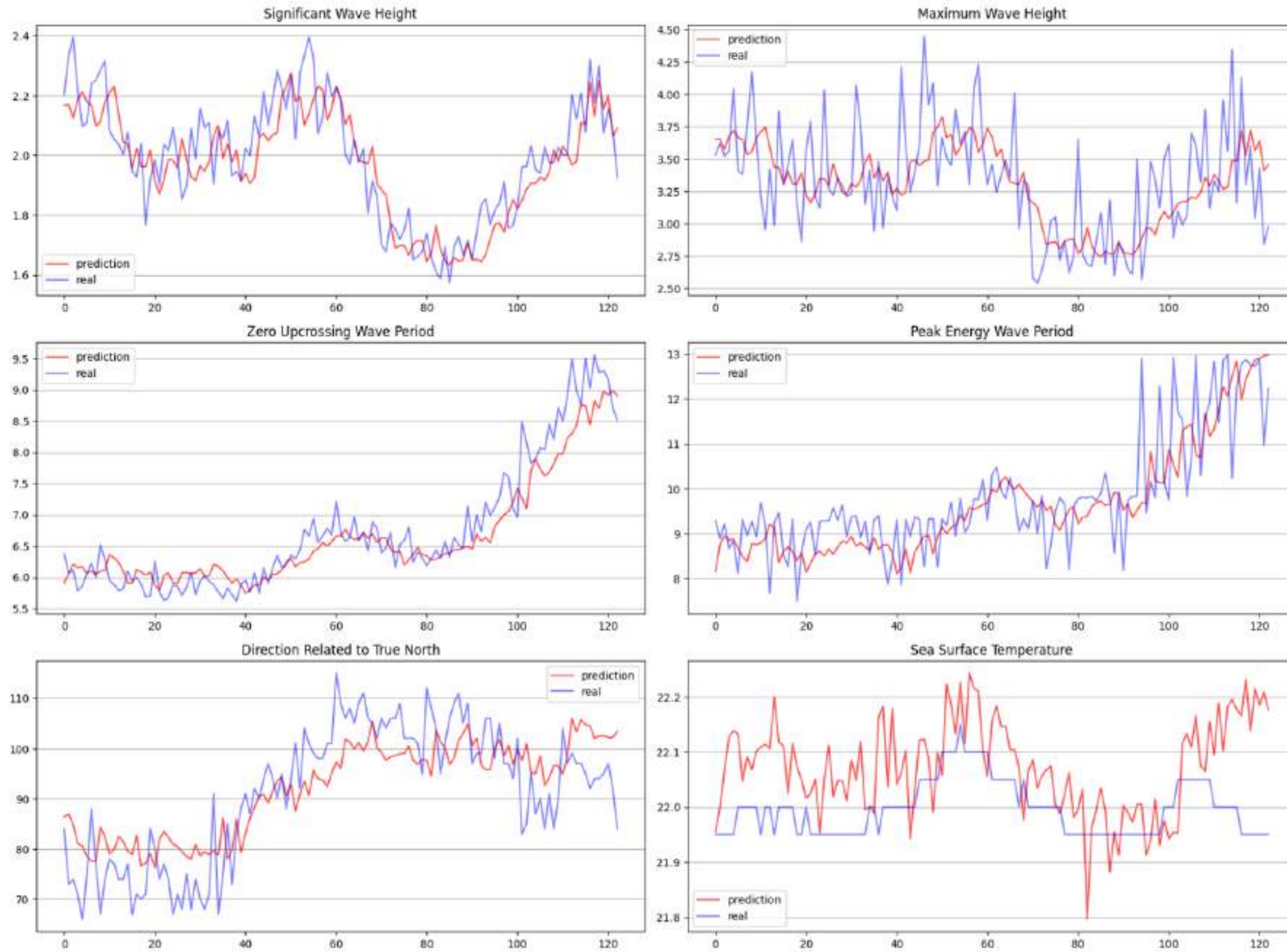
GRU VISUALIZATION: REAL VS PREDICTED



TFT

- The TFT model was trained using processed multivariate time-series data for
- 50 epochs with Mean Squared Error (MSE) as the loss function and Adam optimizer (learning rate = 0.001).
- A ReduceLROnPlateau scheduler was applied to lessen the learning rate based on validation loss, and gradient clipping was carried out to stabilize training.
- Model performance was monitored on the validation set with early stopping.

TFT VISUALIZATION: REAL VS PREDICTED



RESULTS

- In **RMSE** analysis, TFT has resulted in better performance than both LSTM and GRU across all six parameters.
- The **MAE** analysis was also performed across all three models. Out of 6, in 5 parameters, TFT achieved better performance than others. Only for Peak Energy Wave Period (T_p), LSTM marginally outperformed TFT.
- **MAPE** calculates the average error in percentage which is helpful to interpret across different scale, achieved a similar result to MAE with TFT overall the best performing model.
- R^2 score of TFT was calculated as 0.9797, which was the highest, followed by LSTM (0.9774) and GRU (0.9606).

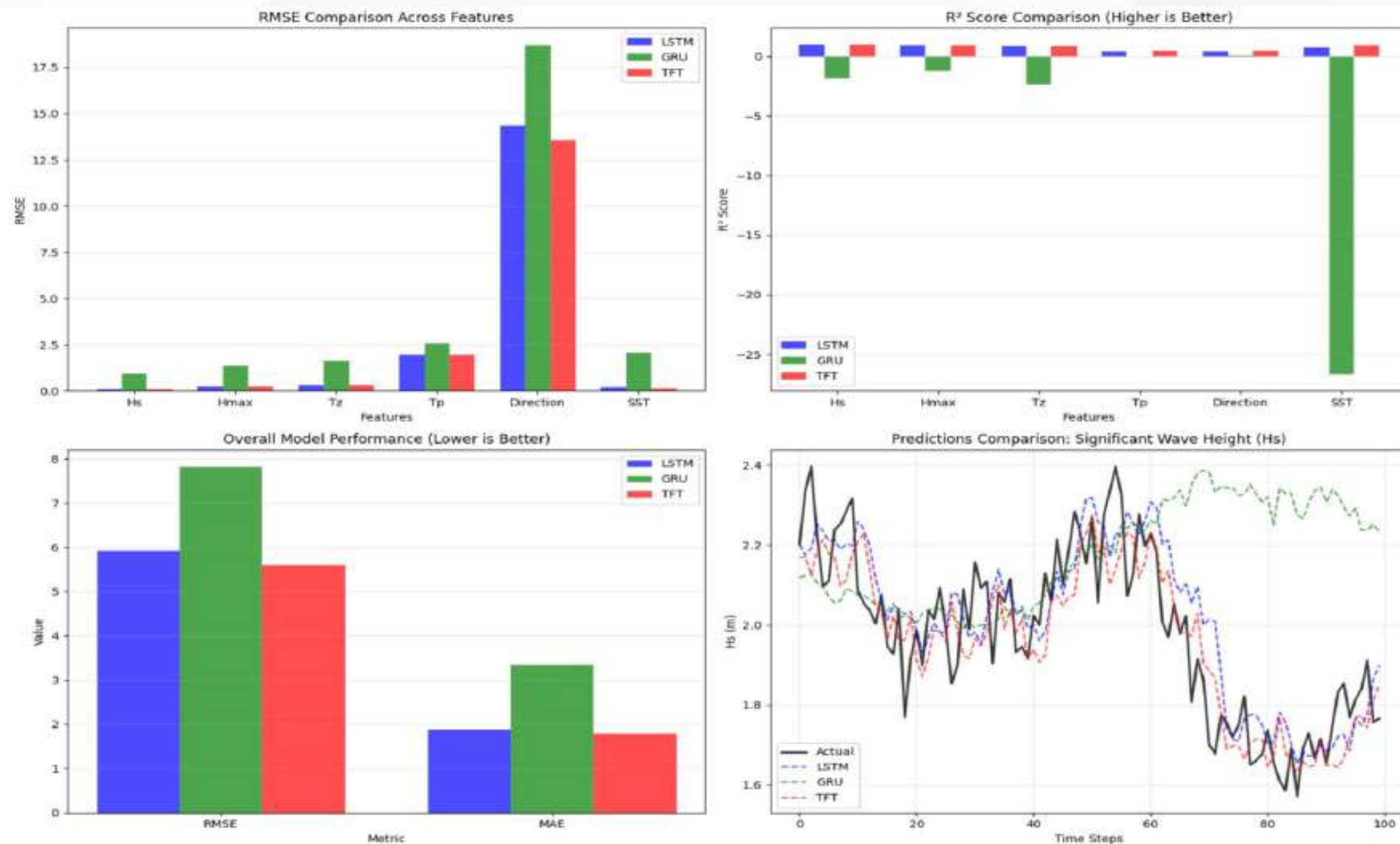
Model Comparison with RMSE

Feature	LSTM	GRU	TFT	Best Model
Hs	0.0889	0.9093	0.0824	TFT
Hmax	0.2484	1.3672	0.242	TFT
Tz	0.3416	1.6312	0.3194	TFT
Tp	1.9561	2.5959	1.9391	TFT
Direction	14.3529	18.6922	13.5743	TFT
SST	0.2117	2.0568	0.1148	TFT
Overall	5.917	7.8073	5.6006	TFT

Model Comparison with MAE

Feature	LSTM	GRU	TFT	Best Model
Hs	0.0673	0.8256	0.0611	TFT
Hmax	0.182	1.238	0.1723	TFT
Tz	0.2595	1.4075	0.2365	TFT
Tp	1.4329	2.1708	1.4345	LSTM
Direction	9.1924	12.4436	8.7315	TFT
SST	0.1528	1.9832	0.0893	TFT
Overall	1.8812	3.3448	1.7875	TFT

Graphical representation also reveals that while LSTM and TFT prediction paths stay closer to the actual track, GRU exhibits larger deviation, particularly during rapid fluctuations.



Model Performance Comparison: LSTM vs GRU vs TFT

CONCLUSION

This research work has provided a detailed comparative analysis of three deep learning models: LSTM, GRU, TFT. TFT achieves its better performance from its attention-based architecture. Though **TFT** provides better performance, its computational complexity against deployment constraints should also be considered. In a situation where resources are limited, LSTM may give a balanced result of efficiency and accuracy.

FUTURE ENHANCEMENT

- Extending the prediction horizon to multi-step forecasting.
- Adding other meteorological variable in the model.
- Addressing special maritime situation like cyclone, earthquake will further increase reliability of the model

OUTCOME



4th International Conference on
Inventive Computing and Informatics
ICICI 2026

10-12, June 2026 - <https://icinci.com/2026/> - iciciconfcontact@gmail.com

Acceptance Letter

Paper Title: Multi-Target Ocean Wave Forecasting Using Deep Learning: Comparative Analysis of Recurrent and Transformer-Based Architectures

Acceptance id: ICICI-035

Authors: Nviz Yadav, Aritra Nanda, Saptak Singha, Mehfooza M

Dear Authors,

On behalf of the Conference committee, we would like to congratulate you on your article to the **ICICI 2026** IEEE Conference, which will be held from **10-12, June 2026** at **S.E.A College of Engineering and Technology**, Bengaluru, Karnataka, India. You have been selected to deliver your presentation at the 4th International Conference on Inventive Computing and Informatics - ICICI 2026.

As a result of the review and results, we are pleased inform that you can now submit the camera-ready paper for inclusion into the ICICI proceedings. We appreciate if you could send the final version of your research paper at your earliest convenience, in order to ensure the timely publication. When submitting your final paper, please highlight the changes made according to the review comments.

Best Regards,

Dr. Usha Desai
Organizing Secretary
ICICI 2026

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The background features a large, light blue diamond shape in the center. The corners are decorated with abstract geometric patterns of blue and white shapes, including hexagons and triangles.

THANK YOU